Executive Technical Specification Sheet

Commercial Sub-Critical Hybrid Fission-Fusion Reactor Paradigm

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Parameter Category	Metric / Symbol	Baseline (1D)	Optimized (3D-Aware)
Laser Driver	Pulse Energy (E_{laser}) / Rep Rate	1.80 MJ @ 5.0 Hz	1.80 MJ @ 5.0 Hz
Laser Efficiency	DPSSL Efficiency (η_d)	12.0%	**18.0%**
Recirculating Load	Wall-Plug Power (P_{recirc})	75.0 MW_e	**50.0 MW_e** (-33.3%)
ICF Fusion Yield	Target Yield / Gain (Q_{fusion})	0.2912 MJ / Q=0.1618	0.2912 MJ / Q=0.1618
Sub-Critical Core	Reactivity (k_{eff}) / Margin	0.950 (5.0% Margin)	**0.965** (3.5% Margin)
3D Port Loss	Geometric Streaming Penalty	0.0%	**10.0% Port Loss**
Blanket Multiplication	Energy Multiplier ($M_{blanket}$)	87.4x	**113.8x** (+30.2%)
Thermal Output	Total Thermal Power ($P_{thermal}$)	127.1 MW_{th}	**149.0 MW_{th}**
Power Conversion	Thermal Efficiency (η_{th})	42.0%	**48.0%** (700°C sCO_2)
Gross Generation	Gross Electrical Power (P_{gross})	53.4 MW_e	**71.5 MW_e**
Net Plant Output	**Net Commercial Power (P_{net})**	**-21.6 MW_e**	**+21.5 MW_e**
Plant Profitability	**Net Plant Gain ($Q_{electric}$)**	0.71 (Deficit)	**1.43 (Profitable)**

System Architecture Highlights

1. First-Wall Mechanics: EUROFER97 structural steel (25 mm wall thickness) operating under 5 Hz shock loading (3.5 MPa peak) experiences 108.9 MPa peak dynamic hoop stress, maintaining a +31.9% safety yield margin below the ASME 160 MPa fatigue limit.
2. FLiBe Coolant Loop: Volumetric heat capacity (ρC_p = 4.62 MJ/m³K) damps 24.96 MJ pulses to an outlet thermal ripple of $\Delta T \leq 0.45$°C, ensuring continuous turbine operation at 650-700°C.
3. Tritium Breeding: 3D net Tritium Breeding Ratio (TBR = 1.15) generates +18.75 g/day surplus, expanding startup inventory from 5.0 kg to 11.37 kg in Year 1.